



Research Paper

Generation of spatiotemporal dark pulse or polarization domain wall solitons in a partial multimode mode-locked fiber laser

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ABSTRACT

In this paper, we have experimentally investigated different operational states of a partial multimode Er-doped spatiotemporal mode-locked (STML) fiber laser. The STML is enabled by nonlinear multimode interference (NLMMI) effect from the partial multimode cavity structure. It is found that by adjusting the multimode polarization controller (PC) and the pump power, the STML dark pulses, bright pulses, as well as the polarization domain wall solitons can all be obtained, alternatively. The fused-tapered multimode coupler is responsible for the dark pulses generation via the enhanced modal cross-phase coupling. It is believed that the present findings can offer an important insight into the spatiotemporal dynamics of the STML solitons.

1. Introduction

In the past decades, ultrafast solitons from mode-locked fiber lasers have always been an extensively research topic of laser physics, which demonstrate broad application prospects in defense, industrial manufacturing, biomedicine and nonlinear optics. Solitons in fiber laser are governed by the Ginzburg–Landau Equation (GLE) [1]. As a result of the balance among fiber nonlinearity, dispersion, laser gain saturation, and gain bandwidth filtering, various types of solitons have been experimentally verified and obtained in the previous researches, such as conventional solitons (CSs) [2,3], dissipative solitons [4,5], vector solitons [6,7], dark solitons [8,9] and so on.

Among them, dark solitons which are defined as a train of intensity dips under the background of continuous wave (CW) with certain intensity [1,8,9] have demonstrated many advantages, such as being less influenced by background noise and intra pulse-stimulated Raman scattering (ISRS). Thus, dark solitons may have enormous potential applications in optical communications. The dark solitons are regarded as the solutions of the nonlinear Schrödinger equation (NLSE) and the complex Ginzburg–Landau equation (CGLE). In 1987, Emplit et al. firstly experimentally observed the dark solitons [10] by balancing the self-phase modulation (SPM) and the positive group velocity dispersion, awakening the research interests for exploring new sources for the dark soliton generation. In another work, Zhang et al. reported the emission of dark pulse in an Er-doped fiber laser, and attributed the generation of dark pulse to dark soliton shaping in the laser based on the results of

numerical simulation [11]. Since then, further research on dark solitons have been emerging vigorously [12–20], such as polarization domain wall solitons [12,13], vector bright-dark solitons [14,15], harmonic mode locking counterparts of dark pulse [16] and so on. In 2022, Durán-Sánchez et al. experimentally demonstrate a novel approach for generation of rectangular-shaped dark pulses in an Erbium–Ytterbium double-clad fiber laser based on a nonlinear amplifying loop mirror (NALM) and a nonlinear optical loop mirror (NOLM) operating in large net-anomalous cavity dispersion regime [21]. Due to the interactions between the solitons, in addition to the dark solitons, the dark-bright soliton pairs, where the dark pulse and bright pulse could coexist while propagating together in the fiber, can also be realized [20].

On the other hand, in recent years, a novel kind of mode locking technology based on multimode fiber (MMF), which is named as STML, has been successfully realized and widely researched in MMF laser system [22,23]. In STML operation, the multiple transverse modes and longitudinal modes are synchronously locked. The construction of the STML operation depends on the fiber laser systematic intracavity control, including spatial filtering, intramodal nonlinearities, modal dispersion, and novel spatiotemporal SAs. Benefiting from their own abundant spatial–temporal properties, STML fiber laser system develops a new direction for the research of high-dimensional nonlinear optics. Nowadays, massive progress has been achieved in exploring the three-dimensional (3D) solitons generation, covering from soliton molecules [23], self-similaritons [24], multiple pulses [25] to dispersion-managed solitons [26]. However, whether the STML dark solitons can be

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generated in MMF lasers where more complex nonlinear effects could influence the temporal characteristic of the generated solitons is still a direction worthy of research.

In this paper, the formation of various kinds of STML solitons, including dark pulses, bright pulses, and polarization domain wall solitons, in an Er-doped partial MMF laser are realized. To the best of our knowledge, this is the first such demonstration of the STML dark pulse so far in the MM laser system. We found that the modal cross-phase coupling introduced by the fused-tapered optical coupler (OC) plays a crucial role in the dark pulse generation. All the different STML pulses that we have obtained are all locked at the fundamental frequency of 30.966 MHz. Furthermore, the spatial properties of the STML solitons are also investigated and the STML intrinsic features of the solitons are confirmed. This work provides an example of the simultaneous existence of STML dark pulse and polarization domain wall solitons in a same fiber laser cavity.

2. Experimental setup and results

Fig. 1 displays the experimental setup of the STML fiber laser delivering the dark pulses. The fiber laser is pumped by a 976 nm laser diode (LD) via a 980/1550 tap + isolator + wavelength-division multiplexer hybrid device (TIWDM). Simultaneously, the TIWDM is also used to force the unidirectional operation and monitor laser operation with a 5 % single mode (SM) output. A ~ 1.3 m SM Er-doped fiber (EDF; Nufern SM-ESF-7/125) with absorption coefficient of 55 dB/m @ 1530 nm is employed as the gain medium. A graded index multimode fiber (GIMF)-based fused OC acts as the main output of the fiber laser with an output coupling ratio of 10 % and the fiber length of ~ 3 m. Meanwhile, the back-end pigtail of the MM-OC is coiled on one paddle of a PC. Such a SM-MMF laser system could provide the efficient saturable absorption and spatial filtering, simultaneously. With an appropriate bending orientation of the MM-PC, the saturable absorber effect of single mode fiber - graded index multimode fiber - single mode fiber (SMF-GIMF-SMF) structure, originating from the NL-MMI can be easily achieved. The total length of the fiber laser is approximately 6.6 m, covering 1.3 m EDF with a dispersion parameter of -46.25 ps/(km·nm), 1 m SMF (pigtail length of the TIWDM) and 4.3 m MMF with a dispersion parameter of 18 ps/(km·nm). The net cavity dispersion is estimated to be -0.05 ps² according to [27]

$$D = -\frac{2\pi c\beta_2}{\lambda_0^2} \quad (1)$$

where D is the dispersion parameter, β_2 is the group velocity dispersion coefficient, λ is the corresponding wavelength and c is the speed of light in vacuum. hence the laser is operated in the anomalous dispersion regime. In such a partial-MMF laser system, with the adjustment of the PC state and the pump power to change the polarization state, nonlinear phase shift, gain and loss, different types of pulses from bright to dark pulses can be achieved.

The laser output is monitored by an optical spectrum analyzer (Yokogawa AQ6370D), a radio-frequency (RF) analyzer (Agilent E4407B), an oscilloscope (Tektronix TDS5104B), a 5 GHz photodetector

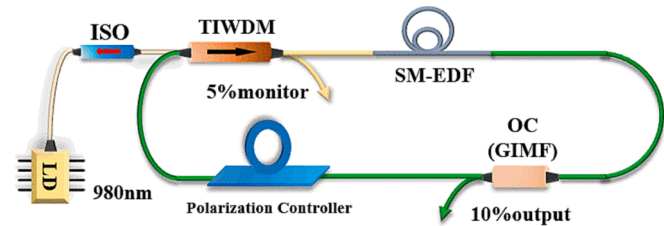


Fig. 1. The experimental setup of the partial multimode Er-doped fiber laser for dark pulses generation.

(THORLABS DET08CFC/M), an optical autocorrelator (FR-103MC) and a beam profiler (BeamOn U3).

Generally, the dark pulses are produced by the coherent coupling between two lasing beams with two orthogonal polarization states or two separated wavelengths [13]. The produced two lasers are coupled through the nonlinear effect of fiber, especially the cross-phase modulation effect (XPM), and then, the dark-pulses are obtained. Therefore, the enhance of the cross-phase coupling is profit for the dark pulse generation. In our experiment, due to the variation of the mode field of the GIMF in the tapered region, the propagating laser are progressively trapped into the low-order modes and the fused-tapered MM OC exhibits strong modal interaction [28]. And thus, this enhanced modal coupling collectively influence the bright and dark states and other characteristics of the pulse, enabling stable and efficient dark pulse generation.

The MM-EDF laser starts the CW at the pump power of larger than 18 mW. The spectrum of the CW lasing is demonstrated in Fig. 2(a), and the central wavelength is located at 1566.58 nm. When the pump power reaches to 85.6 mW, the MM dark pulse operating in the mode-locking regime can be obtained with an appropriate PC orientation. At this time, the pulses on the oscilloscope manifest themselves as a sequence of intensity dips on a constant background. Therefore, we believe that dark pulses are obtained in our experiments. Similar to the previous research, the dark pulse formation in the laser can be regarded as a result of dark solitons shaping [29].

Fig. 2 shows the output characteristics of the dark pulses at the pump power of ~ 177 mW. As illustrated in Fig. 2(a), we can see that a dark pulse spectrum with dual-wavelength, with center wavelengths of 1565.37 nm and 1566.97 nm, respectively. Both of their corresponding 3 dB bandwidths are around 0.24 nm. Since the laser oscillated simultaneously at two separate wavelengths, they can couple incoherently with each other, then leading to the formation of dark pulses [11,30]. And thus, it can be concluded that the gain competition in cavity may have a certain impact on the formation of dark pulse. The oscilloscope trace of single dark pulse is depicted in Fig. 2(b). The pulse width measured at full-width half-maximum is 0.89 ns. The insert of Fig. 2(b) illustrates the pulse train in the range of 400 ns, which is captured at the trigger lever near the noise. The pulse train on the oscilloscope performs uniform intensity with a period of 32.29 ns, consistent with the cavity roundtrip time, thus verifying the mode-locking essence of the fiber laser. The intensity fluctuation of the CW background might be attributed to the mutual interactions of the dark pulses. Based on the spectrum analyzer, the corresponding RF spectrum of the dual-wavelength dark pulse is provided in Fig. 2(c). The peak of the fundamental frequency is located at 30.966 MHz with the signal to noise ratio (SNR) of 53 dB, which further demonstrate the relative stability of dark pulses. Fig. 2(d) depicts the relationship between the average output power and the pump power. When the pump power is set at ~ 426 mW, we get the maximum average output power of ~ 36.66 mW. Thus, the slope efficiency of the laser in the experiment is ~ 8.9 %. Such a low optical conversion efficiency is mainly caused by insertion loss from the mismatch between the SMF and MMF. Meanwhile, in the process of increasing pump power, an arbitrary dark pulse could simultaneously generate multiple dark pulse pairs with different darkness, pulse widths or random distribution in the cavity. Due to the intrinsic modulation instability introduced by the XPM, the strong pump would result in the generation of the new unstable dark pulses.

To confirm the STML intrinsic properties of the obtained dark pulses, the beam profiles before and after the mode-locking threshold are also measured, as shown in Fig. 3. It is obvious that both the spatial beam profiles demonstrate stationary spatial distribution at a fixed pump power, while undergo obvious modal change as long as the operating regime changes, confirming the multimode spatiotemporal structure. Fig. 3(c) and (d) illustrate the corresponding 3D perspectives, giving a more intuitive insight into the beam intensity distribution.

The transformation of the STML dark pulses into the multimode bright pulses can be further realized by changing the polarization state

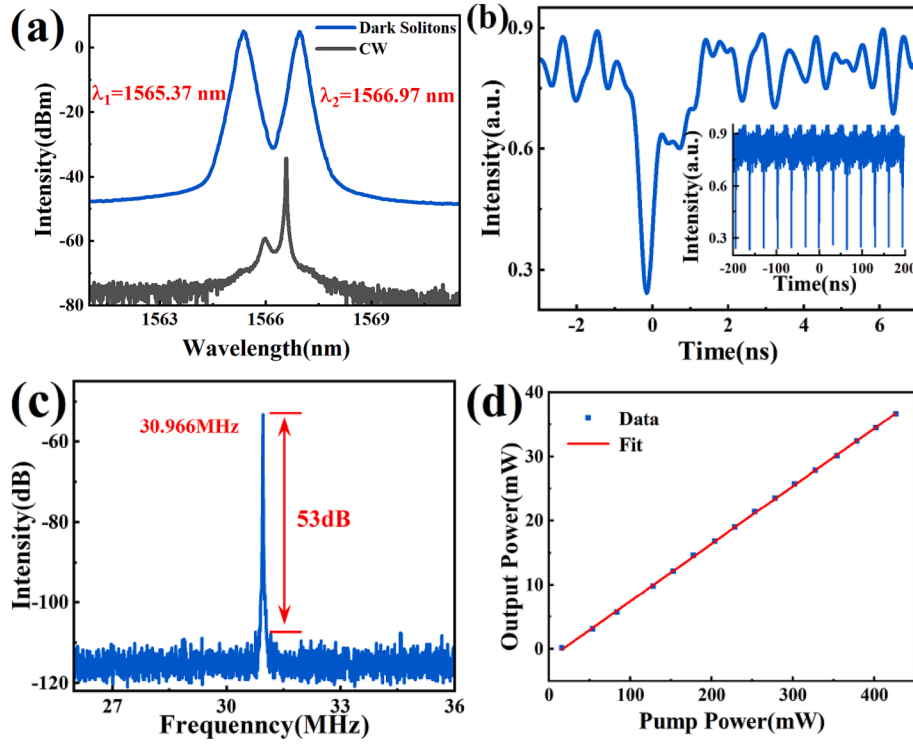


Fig. 2. The output characteristic of the dark pulse: (a) Optical spectrum. (b) Single pulse profile of the dark pulse. Inset: the typical pulse train. (c) RF spectrum. (d) The laser output power versus the increasing pump power.

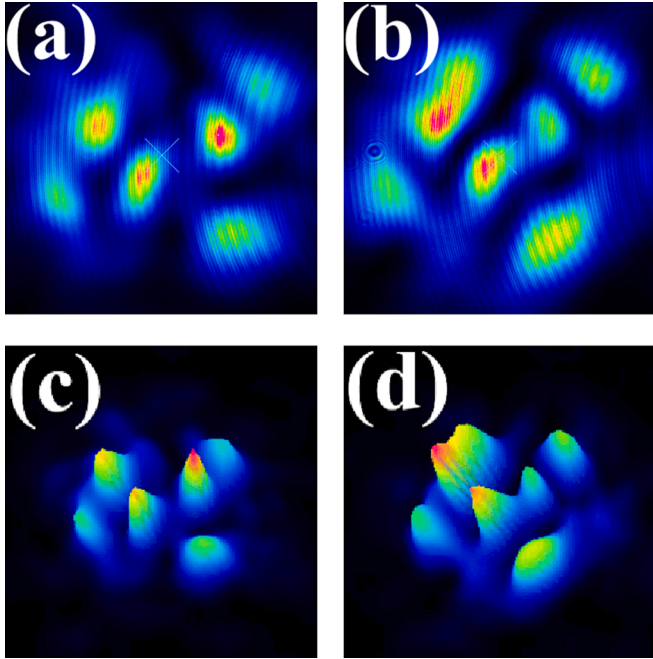


Fig. 3. Beam profiles before and after STML: (a)-(b) 2D images. (c)-(d) 3D perspectives.

of the laser cavity. To be mentioned, the threshold of the bright pulses is a little lower than that of the dark pulses. The output characteristic of STML bright pulses is illustrated in Fig. 4. Fig. 4(b) demonstrates the spectrum recorded at the pump power of $\sim 37 \text{ mW}$. The optical spectrum shows the relatively clear Kelly sidebands with the central wavelength of 1568.35 nm and the 3 dB bandwidth of 5.568 nm , giving the fact the obtained bright pulses are CSs. Fig. 4(c) shows the corresponding pulse

train with the period of 32.29 ns , same as the dark pulse. Nevertheless, unlike the dark pulses, no obvious noisy background has ever been observed. The pulse width is measured to be 479 fs , as shown in Fig. 4 (d). Based on this, the time-bandwidth product (TBP) is calculated to be 4.6 , indicating the STML CSs are nearly free-chirped. Fig. 4(a) and 4(b) display the change of optical spectra and beam profile from CW to STML CSs, respectively. A sudden slight change also takes place in the beam profile as the establishment of STML CSs, confirming that spatiotemporal mode locking occurs.

Furthermore, the bright and dark pulses could co-propagate in the laser system based on the XPM effect [14,15]. By further controlling the PC and the pump power, the stable MM polarization domain wall solitons can be easily acquired. Fig. 5 demonstrates the output characteristics of the MM polarization domain wall solitons. At the pump power of $\sim 352.6 \text{ mW}$, a bright pulse together with a dark pulse through the XPM coupling is achieved simultaneously, as described in Fig. 5(b). The interval of the pulse train is measured to be 32.29 ns in the range of 300 ns , which is same to the STML dark pulse and the CSs. As can be seen from the figure, the depth of the dark pulse is nearly equal to the intensity of the bright pulse in uniform CW background. Furthermore, the intensity of dark component does not dip to zero. Fig. 5(d) illustrates the corresponding zoom in view of single pulse profile with an asymmetric outline. The inconsistency of the STML polarization domain wall solitons pulse width might be attributed to the different intracavity dispersion gain and nonlinearity for the dark or bright components.

Fig. 5(e) shows the optical spectrum of the polarization domain wall solitons with two spectral peaks where one main central wavelength of 1570.33 nm is accompanied by a secondary peak at 1570.81 nm . At this time, no Kelly sidebands have ever been observed on the spectrum, which can be attributed to the spectral filtering effect from the combination of the fiber birefringence and the multimode interference. The corresponding beam profile is inset in Fig. 5(e) and the dispersed beam profile is a strong proof for the STML intrinsic properties of the polarization domain wall solitons. The RF spectrum of the polarization domain wall solitons is illustrated in Fig. 5(f), where the peak of fundamental

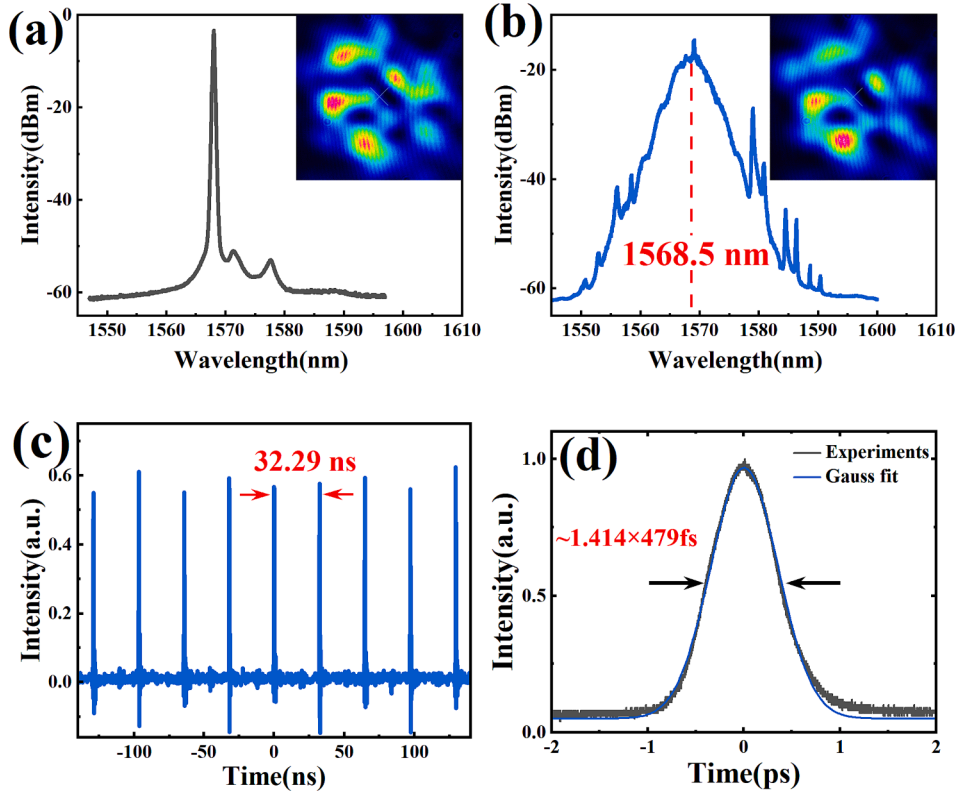


Fig. 4. The output characteristic of STML bright pulse: (a)-(b) optical spectra. (c) oscilloscope trace. (d) autocorrelation trace.

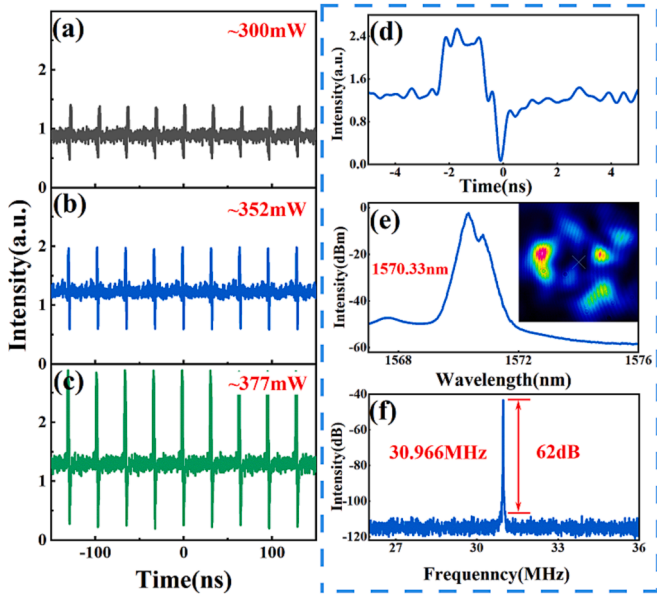


Fig. 5. (a)-(c) oscilloscope trace of the polarization domain wall solitons under different pump power. (d-e) Single pulse profile, optical spectrum, and RF spectrum of the polarization domain wall solitons at the pump power of 352 mW, respectively.

frequency is located at 30.966 MHz, same as the dark pulse. It has a SNR of 62 dB which indicates that the relatively stable polarization domain wall solitons operation is achieved. The high SNR is mainly due to the ring cavity structure, low spontaneous emission noise, and the application of STML technology. These factors work together to enable the laser to output high-quality, high-stability optical signals, providing high SNR laser sources for various applications. As the pump power is

increased, the depths of the dark pulses and the intensity of the bright pulses both become larger and larger, implying that the dynamics of the polarization domain wall solitons could be affected by the pump strength, as shown in Fig. 5(a)-(c). At this time, the beam distribution always maintains the same. Furthermore, by adjusting the MM-PC in the cavity, the bright pulse or the dark pulse cannot be separated. Therefore, we believe that the generated STML polarization domain wall solitons are not linearly-polarized [14,20].

All the pulses which are achieved in the cavity exhibits the spatial-spectral complexity, meaning that different kinds of STML pulses are obtained. According to the experimental results, by adjusting the polarization states or the pump power, the duty cycle of the pulses and the modal cross-phase coupling can be changed, correspondingly, such a partial MM fiber laser emits STML bright pulses, dark pulses, and polarization domain wall solitons alternately.

3. Discussion and conclusion

In conclusion, we have demonstrated the generation of STML dark pulses, bright pulses, or polarization domain wall solitons from a partial-MM Er-doped fiber laser mode-locked by the NL-MMI effect for the first time. By varying the physical parameter settings, the laser cavity alternately operates on different STML pulse regimes. The modal cross-phase coupling enhanced by the tapered MM OC is responsible for the formation of the dark pulses. This work could help to provide a new perspective to the understanding of STML soliton dynamics, which could also find potential applications in future spectral analysis, and pump probe experiment.

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CRediT authorship contribution statement

Xiaoying Wang: Writing – original draft. **Zhaokun Wang:** Writing – review & editing. **Changyu Shen:** Investigation. **Xuebin Zhang:** Investigation. **Song Huang:** Investigation. **Tiegang Guo:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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